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Decorated glass having a printed ink layer

Abstract

Embodiments of a decorated glass are provided. The decorated glass includes a transparent substrate having a first major surface and a second major surface. The second major surface is opposite the first major surface. The decorated glass also includes a black ink layer disposed on the second major surface in a display region. The black ink layer has a transmission coefficient of between 0.2 and 0.85 with respect to incident light having a wavelength of 400 nm to 700 nm. The decorated glass has 2% or less of sparkle when measured from the first major surface via pixel power deviation reference (PPDr).

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
12235477	12/2024	Lesuffleur	N/A	G02B 5/205
2012/0218640	12/2011	Gollier	359/601	C03C 15/00
2012/0300304	12/2011	Gollier	359/599	C03C 15/00
2013/0107370	12/2012	Lander	359/609	C03C 3/087
2016/0169531	12/2015	Wagner et al.	N/A	N/A
2017/0075039	12/2016	Hart et al.	N/A	N/A
2017/0341345	12/2016	Loccufier	N/A	B32B 17/10302
2018/0215657	12/2017	Jin	N/A	C03C 19/00
2019/0235304	12/2018	Tamada	N/A	G02B 5/0221
2019/0248098	12/2018	Ozeki et al.	N/A	N/A
2021/0034100	12/2020	Lesuffleur	N/A	B60K 35/60

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104718071	12/2014	CN	N/A
105693109	12/2015	CN	N/A
105705330	12/2015	CN	N/A
213266287	12/2020	CN	N/A
102016006834	12/2015	DE	N/A
2017/184678	12/2016	WO	N/A
2018/212145	12/2017	WO	N/A

2019/055458	12/2018	WO	N/A

OTHER PUBLICATIONS

Korean Patent Application No. 10-2021-7035648, Notice of Allowance dated Jan. 7, 2025, 3 pages (English Translation only), Korean Patent Office. cited by applicant
Chinese Patent Application No. 202080033703.8, Office Action dated Feb. 13, 2023, 4 pages (English Translation only), Chinese Patent Office. cited by applicant
International Search Report and Written Opinion of the International Searching Authority; PCT/US20/25223; Mailed Jul. 3, 2020; 9 Pages; European Patent Office. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage application under 35 U.S.C. § 371 of International Application No. PCT/US2020/025223, filed on Mar. 27, 2020, which claims the benefit of priority under 35 U.S.C. § 119 of U.S. Provisional Application Ser. No. 62/987,563 filed on Mar. 10, 2020 and U.S. Provisional Application Ser. No. 62/829,347 filed on Apr. 4, 2019, the contents of which are relied upon and incorporated herein by reference in their entirety.

BACKGROUND
(1) The disclosure relates to a decorated glass, and more particularly to a glass having an ink layer to provide color matching or deadfronting with low sparkle.
SUMMARY
(2) In one aspect, embodiments of the disclosure relate to a decorated glass. The decorated glass includes a transparent substrate having a first major surface and a second major surface. The second major surface is opposite the first major surface. The decorated glass also includes a black ink layer disposed on the second major surface in a display region. The black ink layer has a transmission coefficient of between 0.2 and 0.85 with respect to incident light having a wavelength of 400 nm to 700 nm. The decorated glass has 2% or less of sparkle when measured from the first major surface via pixel power deviation reference (PPDr).
(3) In another aspect, embodiments of the disclosure relate to a method of preparing a decorated glass. In one or more embodiments of the method, a transparent substrate is provided that has a first major surface and a second major surface in which the second major surface is opposite to the first major surface. An ink layer is printed on the second major surface of the transparent substrate in at least one display region. The ink layer has an a* and b* of no more than 5 and an L* of no more than 50 according to the CIE L*a*b* color space. Further, the ink layer has a transmission coefficient of between 0.2 and 0.85 with respect to incident light having a wavelength of 400 nm to 700 nm, and the ink layer has 2% or less sparkle when measured via pixel power deviation reference (PPDr).
(4) In another aspect, embodiments of the disclosure relates to a device. The device includes a decorated glass and a light source. The decorated glass has a first side and a second side with the second side being opposite the first side. Further, the decorated glass includes a transparent substrate having a first major surface and a second major surface with the second major surface being opposite the first major surface. The decorated glass also includes a black ink layer disposed on the second major surface in at least one display region. The ink layer has a transmission coefficient of between 0.2 and 0.85 with respect to incident light having a wavelength of 400 nm to 700 nm. The light source is disposed on the second side of the decorated glass, and 2% or less of

sparkle is measured from the first side of the decorated glass via pixel power deviation reference (PPDr).
(5) Additional features and advantages will be set forth in the detailed description which follows, and in part will be readily apparent to those skilled in the art from that description or recognized by practicing the embodiments as described herein, including the detailed description which follows, the claims, as well as the appended drawings.
(6) It is to be understood that both the foregoing general description and the following detailed description are merely exemplary, and are intended to provide an overview or framework to understanding the nature and character of the claims. The accompanying drawings are included to provide a further understanding, and are incorporated in and constitute a part of this specification. The drawings illustrate one or more embodiment(s), and together with the description serve to explain principles and operation of the various embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** depicts a partial cross-sectional view of an electronic device, according to an exemplary embodiment.
(2) FIG. **2** depicts a cross-sectional view of the layers of a decorated glass, according to an exemplary embodiment.
(3) FIG. **3** is a graph of the transmittance for various dilutions of the black in layer, according to an exemplary embodiment;
(4) FIG. **4** depicts an experimental setup for measuring the pixel power deviation reference (PPDr), according to an exemplary embodiment.
(5) FIG. **5** is a side view of a curved deadfront for use with a display, according to an exemplary embodiment.
(6) FIG. **6** is a front perspective view of a decorated glass of FIG. **5** prior to curve formation, according to an exemplary embodiment.
(7) FIG. **7** shows a curved decorated glass shaped to conform to a curved display frame, according to an exemplary embodiment.
(8) FIG. **8** shows a process for cold forming a decorated glass to a curved shape, according to an exemplary embodiment.
(9) FIG. **9** shows a process for forming a curved decorated glass utilizing a curved glass layer, according to an exemplary embodiment.
DETAILED DESCRIPTION
(10) Referring generally to the figures, various embodiments of a decorated glass with an ink layer are provided. In embodiments, the decorated glass may be a cover glass that provides color matching or deadfronting for a display, or in embodiments, the decorated glass may be a window, door, or other architectural panel that obscures a view through the decorated glass from a side of the decorated glass upon which ambient light is incident. Referring to the cover glass embodiment, a decorated glass may be placed over a light source, such as a display, to hide or diminish the visibility of a display when the display is off. A color matching decorated cover glass will not completely hide a display when the display is off, but it obscures the display to make it less prominent. A deadfront decorated cover glass is designed to completely hide the display when the display is off. Both of a color matching decorated cover glass and a deadfront decorated cover glass are designed to allow the display to be visible when the display is on. In general, the difference between a color matching decorated cover glass and a deadfront decorated cover glass is the level of transparency of the color matching or deadfronting layer on the decorated glass. In the present disclosure, the color matching or deadfronting layer is an ink layer printed on the decorated glass.

However, depending on the ink used and how it is applied, there may be unsmooth coverage, surface tension fluctuations, particles, etc. that allow light reflected from the display to leak through the ink layer. According to the present disclosure, the ink layer is applied such that the sparkle caused by this light leakage is less than 2% as measured by pixel power deviation reference (PPDr). Embodiments of the decorated glass discussed herein are provided by way of example and not by way of limitation.
(11) FIG. **1** is a partial cross-sectional view of an electronic device **100** including a touch interface **102**. In embodiments, the electronic device **100** is a standalone device, such as a laptop computer, a tablet computer, a smart-phone, a digital music player, portable gaming station, a television, etc. That is, a "standalone electronic device" **100** is primarily a display screen or interactive panel not incorporated into another structure, device, or apparatus. In other embodiments, the electronic device **100** is incorporated into another structure, device, or apparatus. For example, the electronic device **100** may be a control panel, e.g., in a vehicle, on an appliance, for an elevator, etc., that allows for interaction with the structure, device, or apparatus.
(12) In the embodiment depicted in FIG. **1**, the electronic device **100** includes the touch interface **102**, a housing **104**, a decorated glass **106**, a light source (e.g., display unit **108**), and a circuit board **110**. The housing **104** at least partially surrounds the touch interface **102**, and in the embodiment depicted, provides a seating surface **112** for the decorated glass **106**. Thus, in the embodiment depicted, the decorated glass **106** operates as a cover glass for the display unit **108**. Further, in a standalone device, the housing **104** may provide the boundaries of the electronic device **100**, whereas when the electronic device **100** is incorporated into another structure, device, or apparatus, the housing **104** may just provide a mount for the electronic device **100** within the larger overall structure, device, or apparatus. In either configuration, the decorated glass **106** covers at least a portion of the touch interface **102** and may be seated into the housing **104** to as to provide a substantially planar viewing surface **114**. The circuit board **110** supplies power to the touch interface **102** and to the display unit **108** and processes inputs from the touch interface **102** to produce a corresponding response on the display unit **108**.
(13) The touch interface **102** may include one or more touch sensors in order to detect one or more touch or capacitive inputs, such as due to the placement of a user's finger, stylus, or other interaction device close to or on the decorated glass **106**. The touch interface **102** may generally be any type of interface configured to detect changes in capacitance or other electrical parameters that may be correlated to a user input. The touch interface **102** may be operably connected to and/or in communication the circuit board **110**. The touch interface **102** is configured to receive inputs from an object (e.g., location information based on a user's finger or data from the input device). The display unit **108** is configured to display one or more output images, graphics, icons, and/or videos for the electronic device **100**. The display unit **108** may be substantially any type of display mechanism, such as an light emitting diode (LED) display, an organic LED (OLED) display, a liquid crystal display (LCD), plasma display, or the like.
(14) As mentioned above, the display unit **108** has an internal reflectivity based on the construction of the display unit **108**. For example, a direct-lit backlight LCD display unit **108** may contain several layers in front of the light source, such as a polarizers, glass layers, thin film transistor, liquid crystal, color filter, etc. that internally reflect some of the light incident upon the decorated glass. As also mentioned above, this light may leak through a cover matching or deadfront decorated glass **106** if the color matching or deadfronting layer contains defects.
(15) As mentioned, in other embodiments, the decorated glass **106** may be a window, door, or other architectural, structural, or decorative panel. For example, the decorated glass **106** may be the window of a vehicle or of a building. In such embodiments, the decorated glass **106** may provide privacy or shading for people or objects on the opposite side of the decorated glass **106** from an ambient light source. Additionally, the decorated glass **106** can be a panel in a vehicle, such as a dashboard panel, arm rest, or center console panel. In such embodiments, the decorated glass **106**

may hide, e.g., wires and structural components of the vehicle and provide an overall decorative surface.

(16) Having described the general structure of an exemplary electronic device **100**, the structure of the decorated glass **106** is now described. As can be seen in FIG. **2**, the decorated glass **106** includes a substrate **120**, a black ink layer **122**, and optionally an opaque mask layer **124** (e.g., when used as a cover glass). In embodiments, the substrate **120** is a transparent glass. For example, suitable glass substrates **120** may include at least one of silicates, borosilicates, aluminosilicates, aluminoborosilicates, alkali aluminosilicates, and alkaline earth aluminosilicates, among others. Such glasses may be chemically or thermally strengthened, and embodiments of such glasses are provided below. In embodiments, the substrate **120** has a thickness (i.e., distance between a first major surface **126** and a second major surface **128**) of no more than about 2 mm, no more than about 0.8 mm, or no more than about 0.55 mm.

(17) In embodiments, the black ink layer **122** and the opaque mask layer **124** are applied to the glass substrate **120** in such a way as to define a display region **130** and a non-display region **132**. In particular, portions of the decorated glass **106** that include the opaque mask layer **124** define the non-display region **132**, and regions containing only the black ink layer **122** define the display region **130**. The black ink layer **122** is applied to the first major surface **126** of the glass substrate **120** in display regions **130**. In embodiments that do not include an opaque mask layer **124**, the entire decorated glass **106** is a display region **130**. In non-display regions **132**, the opaque mask layer **124** may be applied to the first major surface **126** or over the black ink layer **122**. For example, a black ink layer **122** may be applied over all or a substantial portion of the first major surface **126**, and the opaque mask layer **124** may be applied over the black ink layer **122** to define the non-display regions **132**. In an alternative example, the opaque mask layer **124** may be applied first to the first major surface **126** to create the non-display regions **132**, and the black ink layer **122** may be applied only in the display regions **130** (i.e., where there is no opaque mask layer **124**) or over all or a portion of the opaque mask layer **124**.

(18) In order to provide the color matching or deadfront effect, the black ink layer **122** in the display regions **130** has a transmission coefficient of 0.2 to 0.85 for light incident on the second major surface **128** of the substrate **120**. That is, the black ink layer **122** transmits between 20% and 85% of light in the visible spectrum (i.e., light having a wavelength of 400 nm to 700 nm) that is incident upon the second major surface **128**. That is, the black ink layer **122** has a transmission coefficient of 0.2 to 0.85. In other embodiments, the black ink layer **122** has a transmission coefficient of 0.2 to 0.8, and in still other embodiments, the black ink layer **122** has a transmission coefficient of 0.2 to 0.75. In embodiments, the black ink layer **122** will provide a color matching effect if the transmission coefficient is from 0.5 to 0.85. In embodiments, the black ink layer **122** will provide a deadfront effect if the transmission coefficient is from 0.2 to 0.7. As can be seen, there is an overlap between the transmission coefficient for color matching and deadfronting because the visibility of an object behind the decorated glass **106** depends on certain characterisitics of the object, such as the reflectivity of the object. Thus, an object the reflects less light may achieve a deadfront effect at a higher transmission coefficient, whereas an object that reflects more light may require a lower transmission coefficient just to achieve color matching.

(19) In embodiments, the black ink layer **122** is printed onto the first major surface **126** (or opaque mask layer **124**). In embodiments, the black ink layer **122** is printed via inkjet printing, slot printing, screen printing, pad printing, or gravure printing, among others. In embodiments, the black ink layer **122** is has a thickness of no more than 50 μm. In other embodiments, the black ink layer **122** has a thickness of no more than 30 μm, and in still other embodiments, the black ink layer **122** has a thickness of no more than 20 μm. In embodiments, the black ink layer **122** has a thickness of at least 0.05 μm.

(20) Further, the ink of the black ink layer **122** is selected and printed in such a way that the black ink layer **122** is neutral density (i.e., has no color). With respect to the CIE L*a*b* color space, a*

and b* are no more than 5 for the black ink layer **122**. In embodiments, a* and b* are no more than 2, and in still other embodiments, a* and b* are no more than 1. In particular embodiments, a* and b* are 0. In embodiments, L* is less than 50. In other embodiments, L* is less than 30, and in still other embodiments, L* is less than 20. In embodiments, the black ink layer **122** comprises dyes and/or pigments, such as carbon black. Further, in embodiments, the black ink layer **122** is CMYK composite black (i.e., a mixture of cyan, magenta, and yellow ink). In other embodiments, the black ink layer **122** is a CMYK rich black (i.e., a mixture of cyan, magenta, yellow, and black ink). In still other embodiments, the black ink layer **122** is printed using just K (black) ink or LK (light black) ink according to CMYK. In embodiments, the transmittance of the black ink layer **122** is controlled by diluting the ink with solvent. Specifically, a more dilute ink will produce a black ink layer **122** having a higher transmittance that a less dilute ink. FIG. **3** provides a graph of the transmittance for four inks that have been diluted with 2%, 4%, 6%, and 8% (by volume) with a solvent. As can be seen in FIG. **3**, the 2% diluted ink had the lowest transmittance in the visible spectrum, and the 8% diluted ink had the highest transmittance in the visible spectrum. Thus, for example, the 2% diluted ink could be used to provide a deadfront effect, whereas the 8% diluted ink could be used to provide a color matching effect.

(21) Additionally, printing at higher ink volumes (i.e., less diluted) reduced the observable sparkle of the decorated glass **106**. FIG. **4** depicts an experimental setup to measure sparkle of the decorated glass **106** via PPDr. The image system comprises a high-resolution CCD camera, imaging lenses L**1** and L**2**, and a diaphragm D. The imaging lenses L**1**, L**2** are chosen to achieve the desired ratio of CCD camera pixels to source (i.e., display unit **108**) pixels. The diaphragm D is set to simulate the collection angle of the human eye, e.g., about 12 mrad to 18 mrad. A pixel power deviation (PPD) was calculated for a reference transparent substrate without black ink layer **122**, which then served as the reference to the PPDr for the decorated glass **106** having the black ink layer **122** printed thereon. A sparkle value was calculated for each of the decorated glasses **106** shown in FIG. **3**. Going from least diluted to most diluted, the sparkle measures were 1.06%, 1.30%, 2.07%, and 1.96%. In general, Applicant found that the sparkle measurements were lowest for the least diluted inks. In embodiments, the ink used to print the ink layer **124** is diluted by no more than 10% by volume with a solvent. Further, in embodiments, the sparkle of the decorated glass **106** is no more than 2% as measured via PPDr. In other embodiments, the sparkle of the decorated glass **106** is no more than 1.5%, and in still other embodiments, the sparkle of the decorated glass **106** is no more than 1.25%.

(22) As discussed above, the decorated glass **106** may also include an opaque mask layer **124**, especially when used as a cover glass. The opaque mask layer **124** transmits less than 5% of incident light. In particular, the opaque mask layer **124** transmits less than 0.5% of incident light. In this way, the opaque mask layer **124** blocks visibility of any components beneath the decorated glass **106** in the non-display regions **132**. For example, the opaque mask layer **124** may be used to block visibility of connections to the display unit **108** below the decorated glass **106**, a border of the display unit **108**, circuitry, etc. In embodiments, the opaque mask layer **124** is selected to have an optical density of at least 3. The opaque mask layer **124** may be applied using screen printing, inject printing, spin coating, and various lithographic techniques, among others. In embodiments, the opaque mask layer **124** has a thickness of from 1 μm to 20 μm. In embodiments, the opaque mask layer **124** is selected to be gray or black in color; however, other colors are also possible depending on the need to match any other colors in the decorated glass **106**.

(23) Additionally, as depicted in FIG. **1**, the decorated glass **106** can include a surface treatment **134** on one or both of the first major surface **126** and the second major surface **128**. The surface treatment **106** can be provided through addition or removal of material from the first or second major surface **126**, **128**. For example, the surface treatment **134** can be applied by coating the first or second major surface **126**, **128**. In another example, the surface treatment **134** can be a removal of material from the first or second major surface **126**, **128** such as through etching. Exemplary

surface treatments include anti-fingerprint, anti-reflection, and anti-glare. In an embodiment, one or both of the anti-fingerprint and anti-reflection treatments are applied to the second major surface **128**, and the anti-glare treatment is applied to the first major surface **128**.

(24) Embodiments of the decorated glass **106** disclosed herein provide several advantages. For example, as compared to decorated glasses that utilize a film to provide color matching or deadfronting, the decorated glass **106** is easily tunable to different transmittances, e.g., by changing the dilution of the ink or the thickness of the ink layer. Additionally, as compared to conventional films, the decorated glass **106** having the black ink layer **122** does not exhibit any internal reflectance that contributes to sparkle. Indeed, conventional films often include multiple layers, and incident light can reflect off of these layers and contribute to a higher overall sparkle. The black ink layer **122** has no internal layers. Further, the black ink layer **122** is thinner than conventional films, providing a thinner decorated glass **106**.

(25) Further, the decorated glass **106** having the printed black ink layer **122** provides design flexibility. In particular, displays vary between systems, and each display has a particular internal reflectance. Because of the ease by which the transmittance of the black ink layer **122** can be tuned by varying the solvent content, the variance in internal reflectance can quickly and economically be accommodated to provide the desired deadfronting or color matching effect.

(26) Referring to FIGS. **5-9**, various sizes, shapes, curvatures, glass materials, etc. for a decorated glass **106** along with various processes for forming a curved decorated glass are shown and described. It should be understood, that while FIGS. **5-9** are described in the context of a simplified curved decorated glass **2000** for ease of explanation, the decorated glass **2000** may be any of the decorated glass embodiments discussed herein.

(27) As shown in FIG. **5**, in one or more embodiments, decorated glass **2000** includes a curved outer glass layer **2010** (e.g., substrate **120**) having at least a first radius of curvature, R**1**, and in various embodiments, curved outer glass layer **2010** is a complex curved sheet of glass material having at least one additional radius of curvature. In various embodiments, R**1** is in a range from about 60 mm to about 1500 mm.

(28) Curved decorated glass **2000** includes a polymer layer **2020** located along an inner, major surface of curved outer glass layer **2010**. Curved decorated glass **2000** also includes a frame **2030** (which may be a metal, plastic, glass, or ceramic material). Still further, curved decorated glass **2000** may also include any of the other layers described above, such as the surface treatment, the opaque mask layer, and an optically clear adhesive to attach a display unit **108** (as shown in FIG. **1**) to the decorated glass **2000**. Additionally, curved decorated glass **2000** may include such layers as, e.g., light guide layers, reflector layers, display module(s), display stack layers, light sources, etc. that otherwise may be associated with an electronic device as discussed herein.

(29) As will be discussed in more detail below, in various embodiments, curved decorated glass **2000**, including glass layer **2010**, polymer layer **2020**, frame **2030**, and any other optional layers may be cold-formed together to a curved shape, as shown in FIG. **5**. In other embodiments, glass layer **2010** may be formed to a curved shape, and then layers **2020** and **2030** are applied following curve formation.

(30) Referring to FIG. **6**, outer glass layer **2010** is shown prior to being formed to the curved shape shown in FIG. **6**. In general, Applicant believes that the articles and processes discussed herein provide high quality decorated glass structures utilizing glass of sizes, shapes, compositions, strengths, etc. not previously provided.

(31) As shown in FIG. **6**, glass layer **2010** includes a first major surface **2050** and a second major surface **2060** opposite first major surface **2050**. An edge surface or minor surface **2070** connects the first major surface **2050** and the second major surface **2060**. Glass layer **2010** has a thickness (t) that is substantially constant and is defined as a distance between the first major surface **2050** and the second major surface **2060**. In some embodiments, the thickness (t) as used herein refers to the maximum thickness of the glass layer **2010**. Glass layer **2010** includes a width (W) defined as a

first maximum dimension of one of the first or second major surfaces orthogonal to the thickness (t), and outer glass layer **2010** also includes a length (L) defined as a second maximum dimension of one of the first or second surfaces orthogonal to both the thickness and the width. In other embodiments, the dimensions discussed herein are average dimensions.

(32) In one or more embodiments, glass layer **2010** has a thickness (t) that is in a range from 0.05 mm to 2 mm. In various embodiments, glass layer **2010** has a thickness (t) that is about 1.5 mm or less. For example, the thickness may be in a range from about 0.1 mm to about 1.5 mm, from about 0.15 mm to about 1.5 mm, from about 0.2 mm to about 1.5 mm, from about 0.25 mm to about 1.5 mm, from about 0.3 mm to about 1.5 mm, from about 0.35 mm to about 1.5 mm, from about 0.4 mm to about 1.5 mm, from about 0.45 mm to about 1.5 mm, from about 0.5 mm to about 1.5 mm, from about 0.55 mm to about 1.5 mm, from about 0.6 mm to about 1.5 mm, from about 0.65 mm to about 1.5 mm, from about 0.7 mm to about 1.5 mm, from about 0.1 mm to about 1.4 mm, from about 0.1 mm to about 1.3 mm, from about 0.1 mm to about 1.2 mm, from about 0.1 mm to about 1.1 mm, from about 0.1 mm to about 1.05 mm, from about 0.1 mm to about 1 mm, from about 0.1 mm to about 0.95 mm, from about 0.1 mm to about 0.9 mm, from about 0.1 mm to about 0.85 mm, from about 0.1 mm to about 0.8 mm, from about 0.1 mm to about 0.75 mm, from about 0.1 mm to about 0.7 mm, from about 0.1 mm to about 0.65 mm, from about 0.1 mm to about 0.6 mm, from about 0.1 mm to about 0.55 mm, from about 0.1 mm to about 0.5 mm, from about 0.1 mm to about 0.4 mm, or from about 0.3 mm to about 0.7 mm.

(33) In one or more embodiments, glass layer **2010** has a width (W) in a range from about 5 cm to about 250 cm, from about 10 cm to about 250 cm, from about 15 cm to about 250 cm, from about 20 cm to about 250 cm, from about 25 cm to about 250 cm, from about 30 cm to about 250 cm, from about 35 cm to about 250 cm, from about 40 cm to about 250 cm, from about 45 cm to about 250 cm, from about 50 cm to about 250 cm, from about 55 cm to about 250 cm, from about 60 cm to about 250 cm, from about 65 cm to about 250 cm, from about 70 cm to about 250 cm, from about 75 cm to about 250 cm, from about 80 cm to about 250 cm, from about 85 cm to about 250 cm, from about 90 cm to about 250 cm, from about 95 cm to about 250 cm, from about 100 cm to about 250 cm, from about 110 cm to about 250 cm, from about 120 cm to about 250 cm, from about 130 cm to about 250 cm, from about 140 cm to about 250 cm, from about 150 cm to about 250 cm, from about 5 cm to about 240 cm, from about 5 cm to about 230 cm, from about 5 cm to about 220 cm, from about 5 cm to about 210 cm, from about 5 cm to about 200 cm, from about 5 cm to about 190 cm, from about 5 cm to about 180 cm, from about 5 cm to about 170 cm, from about 5 cm to about 160 cm, from about 5 cm to about 150 cm, from about 5 cm to about 140 cm, from about 5 cm to about 130 cm, from about 5 cm to about 120 cm, from about 5 cm to about 110 cm, from about 5 cm to about 100 cm, from about 5 cm to about 90 cm, from about 5 cm to about 80 cm, or from about 5 cm to about 75 cm.

(34) In one or more embodiments, glass layer **2010** has a length (L) in a range from about 5 cm to about 250 cm, from about 10 cm to about 250 cm, from about 15 cm to about 250 cm, from about 20 cm to about 250 cm, from about 25 cm to about 250 cm, from about 30 cm to about 250 cm, from about 35 cm to about 250 cm, from about 40 cm to about 250 cm, from about 45 cm to about 250 cm, from about 50 cm to about 250 cm, from about 55 cm to about 250 cm, from about 60 cm to about 250 cm, from about 65 cm to about 250 cm, from about 70 cm to about 250 cm, from about 75 cm to about 250 cm, from about 80 cm to about 250 cm, from about 85 cm to about 250 cm, from about 90 cm to about 250 cm, from about 95 cm to about 250 cm, from about 100 cm to about 250 cm, from about 110 cm to about 250 cm, from about 120 cm to about 250 cm, from about 130 cm to about 250 cm, from about 140 cm to about 250 cm, from about 150 cm to about 250 cm, from about 5 cm to about 240 cm, from about 5 cm to about 230 cm, from about 5 cm to about 220 cm, from about 5 cm to about 210 cm, from about 5 cm to about 200 cm, from about 5 cm to about 190 cm, from about 5 cm to about 180 cm, from about 5 cm to about 170 cm, from about 5 cm to about 160 cm, from about 5 cm to about 150 cm, from about 5 cm to about 140 cm,

from about 5 cm to about 130 cm, from about 5 cm to about 120 cm, from about 5 cm to about 110 cm, from about 5 cm to about 100 cm, from about 5 cm to about 90 cm, from about 5 cm to about 80 cm, or from about 5 cm to about 75 cm.

(35) As shown in FIG. **5**, glass layer **2010** is shaped to a curved shaping having at least one radius of curvature, shown as R**1**. In various embodiments, glass layer **2010** may be shaped to the curved shape via any suitable process, including cold-forming and hot-forming.

(36) In specific embodiments, glass layer **2010** is shaped to the curved shape shown in FIG. **5**, either alone, or following attachment of layers **2020** and **2030**, via a cold-forming process. As used herein, the terms "cold-bent," "cold-bending," "cold-formed" or "cold-forming" refers to curving the glass decorated glass at a cold-form temperature which is less than the softening point of the glass (as described herein). A feature of a cold-formed glass layer is an asymmetric surface compressive between the first major surface **2050** and the second major surface **2060**. In some embodiments, prior to the cold-forming process or being cold-formed, the respective compressive stresses in the first major surface **2050** and the second major surface **2060** are substantially equal.

(37) In some such embodiments in which glass layer **2010** is unstrengthened, the first major surface **2050** and the second major surface **2060** exhibit no appreciable compressive stress, prior to cold-forming. In some such embodiments in which glass layer **2010** is strengthened (as described herein), the first major surface **2050** and the second major surface **2060** exhibit substantially equal compressive stress with respect to one another, prior to cold-forming. In one or more embodiments, after cold-forming the compressive stress on the second major surface **2060** (e.g., the concave surface following bending) increases (i.e., the compressive stress on the second major surface **2050** is greater after cold-forming than before cold-forming).

(38) Without being bound by theory, the cold-forming process increases the compressive stress of the glass article being shaped to compensate for tensile stresses imparted during bending and/or forming operations. In one or more embodiments, the cold-forming process causes the second major surface **2060** to experience compressive stresses, while the first major surface **2050** (e.g., the convex surface following bending) experiences tensile stresses. The tensile stress experienced by surface **2050** following bending results in a net decrease in surface compressive stress, such that the compressive stress in surface **2050** of a strengthened glass sheet following bending is less than the compressive stress in surface **2050** when the glass sheet is flat.

(39) Further, when a strengthened glass sheet is utilized for glass layer **2010**, the first major surface and the second major surface (**2050**, **2060**) are already under compressive stress, and thus first major surface **2050** can experience greater tensile stress during bending without risking fracture. This allows for the strengthened embodiments of glass layer **2010** to conform to more tightly curved surfaces (e.g., shaped to have smaller R**1** values).

(40) In various embodiments, the thickness of glass layer **2010** is tailored to allow glass layer **2010** to be more flexible to achieve the desired radius of curvature. Moreover, a thinner glass layer **2010** may deform more readily, which could potentially compensate for shape mismatches and gaps that may be created by the shape of a support or frame (as discussed below). In one or more embodiments, a thin and strengthened glass layer **2010** exhibits greater flexibility especially during cold-forming. The greater flexibility of the glass articles discussed herein may allow for consistent bend formation without heating.

(41) In various embodiments, glass layer **2010** (and consequently decorated glass **2000**) may have a compound curve including a major radius and a cross curvature. A complexly curved cold-formed glass layer **2010** may have a distinct radius of curvature in two independent directions. According to one or more embodiments, the complexly curved cold-formed glass layer **2010** may thus be characterized as having "cross curvature," where the cold-formed glass layer **2010** is curved along an axis (i.e., a first axis) that is parallel to a given dimension and also curved along an axis (i.e., a second axis) that is perpendicular to the same dimension. The curvature of the cold-formed glass layer **2010** can be even more complex when a significant minimum radius is combined with a

significant cross curvature, and/or depth of bend.
(42) Referring to FIG. **7**, display assembly **2100** is shown according to an exemplary embodiment. In the embodiment shown, display assembly **2100** includes frame **2110** supporting (either directly or indirectly) both a light source, shown as a display unit **2120**, and decorated glass **2000**. As shown in FIG. **7**, decorated glass **2000** and display unit **2120** are coupled to frame **2110**, and display module **2120** is positioned to allow a user to view light, images, etc. generated by display unit **2120** through the decorated glass **2000**. In various embodiments, frame **2110** may be formed from a variety of materials such as plastic (PC/ABS, etc.), metals (Al-alloys, Mg-alloys, Fe-alloys, etc.), glass, or ceramic. Various processes such as casting, machining, stamping, injection molding, etc. may be utilized to form the curved shape of frame **2110**. While frame **2110** is shown as a frame associated with a display assembly, frame **2110** may be any support or frame structure associated with a vehicle interior system.
(43) In various embodiments, the systems and methods described herein allow for formation of decorated glass **2000** to conform to a wide variety of curved shapes that frame **2110** may have. As shown in FIG. **7**, frame **2110** has a support surface **2130** that has a curved shape, and decorated glass structure **2000** is shaped to match the curved shape of support surface **2130**. As will be understood, decorated glass structure **2000** may be shaped into a wide variety of shapes to conform to a desired frame shape of a display assembly **2100**, which in turn may be shaped to fit the shape of a portion of a vehicle interior system, as discussed herein.
(44) In one or more embodiments, decorated glass **2000** (and specifically glass layer **2010**) is shaped to have a first radius of curvature, R**1**, of about 60 mm or greater. For example, R**1** may be in a range from about 60 mm to about 1500 mm, from about 70 mm to about 1500 mm, from about 80 mm to about 1500 mm, from about 90 mm to about 1500 mm, from about 100 mm to about 1500 mm, from about 120 mm to about 1500 mm, from about 140 mm to about 1500 mm, from about 150 mm to about 1500 mm, from about 160 mm to about 1500 mm, from about 180 mm to about 1500 mm, from about 200 mm to about 1500 mm, from about 220 mm to about 1500 mm, from about 240 mm to about 1500 mm, from about 250 mm to about 1500 mm, from about 260 mm to about 1500 mm, from about 270 mm to about 1500 mm, from about 280 mm to about 1500 mm, from about 290 mm to about 1500 mm, from about 300 mm to about 1500 mm, from about 350 mm to about 1500 mm, from about 400 mm to about 1500 mm, from about 450 mm to about 1500 mm, from about 500 mm to about 1500 mm, from about 550 mm to about 1500 mm, from about 600 mm to about 1500 mm, from about 650 mm to about 1500 mm, from about 700 mm to about 1500 mm, from about 750 mm to about 1500 mm, from about 800 mm to about 1500 mm, from about 900 mm to about 1500 mm, from about 9500 mm to about 1500 mm, from about 1000 mm to about 1500 mm, from about 1250 mm to about 1500 mm, from about 60 mm to about 1400 mm, from about 60 mm to about 1300 mm, from about 60 mm to about 1200 mm, from about 60 mm to about 1100 mm, from about 60 mm to about 1000 mm, from about 60 mm to about 950 mm, from about 60 mm to about 900 mm, from about 60 mm to about 850 mm, from about 60 mm to about 800 mm, from about 60 mm to about 750 mm, from about 60 mm to about 700 mm, from about 60 mm to about 650 mm, from about 60 mm to about 600 mm, from about 60 mm to about 550 mm, from about 60 mm to about 500 mm, from about 60 mm to about 450 mm, from about 60 mm to about 400 mm, from about 60 mm to about 350 mm, from about 60 mm to about 300 mm, or from about 60 mm to about 250 mm.
(45) In one or more embodiments, support surface **2130** has a second radius of curvature of about 60 mm or greater. For example, the second radius of curvature of support surface **2130** may be in a range from about 60 mm to about 1500 mm, from about 70 mm to about 1500 mm, from about 80 mm to about 1500 mm, from about 90 mm to about 1500 mm, from about 100 mm to about 1500 mm, from about 120 mm to about 1500 mm, from about 140 mm to about 1500 mm, from about 150 mm to about 1500 mm, from about 160 mm to about 1500 mm, from about 180 mm to about 1500 mm, from about 200 mm to about 1500 mm, from about 220 mm to about 1500 mm, from

about 240 mm to about 1500 mm, from about 250 mm to about 1500 mm, from about 260 mm to about 1500 mm, from about 270 mm to about 1500 mm, from about 280 mm to about 1500 mm, from about 290 mm to about 1500 mm, from about 300 mm to about 1500 mm, from about 350 mm to about 1500 mm, from about 400 mm to about 1500 mm, from about 450 mm to about 1500 mm, from about 500 mm to about 1500 mm, from about 550 mm to about 1500 mm, from about 600 mm to about 1500 mm, from about 650 mm to about 1500 mm, from about 700 mm to about 1500 mm, from about 750 mm to about 1500 mm, from about 800 mm to about 1500 mm, from about 900 mm to about 1500 mm, from about 9500 mm to about 1500 mm, from about 1000 mm to about 1500 mm, from about 1250 mm to about 1500 mm, from about 60 mm to about 1400 mm, from about 60 mm to about 1300 mm, from about 60 mm to about 1200 mm, from about 60 mm to about 1100 mm, from about 60 mm to about 1000 mm, from about 60 mm to about 950 mm, from about 60 mm to about 900 mm, from about 60 mm to about 850 mm, from about 60 mm to about 800 mm, from about 60 mm to about 750 mm, from about 60 mm to about 700 mm, from about 60 mm to about 650 mm, from about 60 mm to about 600 mm, from about 60 mm to about 550 mm, from about 60 mm to about 500 mm, from about 60 mm to about 450 mm, from about 60 mm to about 400 mm, from about 60 mm to about 350 mm, from about 60 mm to about 300 mm, or from about 60 mm to about 250 mm.

(46) In one or more embodiments, decorated glass **2000** is cold-formed to exhibit a first radius curvature, R**1**, that is within 10% (e.g., about 10% or less, about 9% or less, about 8% or less, about 7% or less, about 6% or less, or about 5% or less) of the second radius of curvature of support surface **2130** of frame **2110**. For example, support surface **2130** of frame **2110** exhibits a radius of curvature of 1000 mm, decorated glass **2000** is cold-formed to have a radius of curvature in a range from about 900 mm to about 1100 mm.

(47) In one or more embodiments, first major surface **2050** and/or second major surface **2060** of glass layer **2010** includes a surface treatment or a functional coating. The surface treatment may cover at least a portion of first major surface **2050** and/or second major surface **2060**. Exemplary surface treatments include at least one of a glare reduction coating, an anti-glare coating, a scratch resistance coating, an anti-reflection coating, a half-mirror coating, or easy-to-clean coating.

(48) Referring to FIG. **8**, a method **2200** for forming a display assembly **2100** (as shown in FIG. **7**) that includes a cold-formed decorated glass structure, such as decorated glass **2000**, is shown. At step **2210**, a decorated glass structure, such decorated glass **2000**, is supported and/or placed on a curved support. In general, the curved support may be a frame of a display, such as frame **2110**, that defines a perimeter and curved shape of a vehicle display. In general, the curved frame includes a curved support surface, and one of the major surfaces **2050** and **2060** of decorated glass **2000** is placed into contact with the curved support surface **2130**.

(49) At step **2220**, a force is applied to the decorated glass structure while it is supported by the support causing the decorated glass structure to bend into conformity with the curved shape of the support. In this manner, a curved decorated glass structure **2000**, as shown in FIG. **5**, is formed from a generally flat decorated glass structure. In this arrangement, curving the flat decorated glass forms a curved shape on the major surface facing the support, while also causing a corresponding (but complimentary) curve to form in the major surface opposite of the frame. Applicant believes that by bending the decorated glass structure directly on the curved frame, the need for a separate curved die or mold (typically needed in other glass bending processes) is eliminated. Further, Applicant believes that by shaping the decorated glass directly to the curved frame, a wide range of curved radii may be achieved in a low complexity manufacturing process.

(50) In some embodiments, the force applied in step **2220** may be air pressure applied via a vacuum fixture. In some other embodiments, the air pressure differential is formed by applying a vacuum to an airtight enclosure surrounding the frame and the decorated glass structure. In specific embodiments, the airtight enclosure is a flexible polymer shell, such as a plastic bag or pouch. In other embodiments, the air pressure differential is formed by generating increased air pressure

around the decorated glass and the frame with an overpressure device, such as an autoclave. Applicant has further found that air pressure provides a consistent and highly uniform bending force (as compared to a contact-based bending method) which further leads to a robust manufacturing process. In various embodiments, the air pressure differential is between 0.5 and 1.5 atmospheres of pressure (atm), specifically between 0.7 and 1.1 atm, and more specifically is 0.8 to 1 atm.

(51) At step **2230**, the temperature of the decorated glass structure is maintained below the glass transition temperature of the material of the outer glass layer during the bending. As such, method **2200** is a cold-forming or cold-bending process. In particular embodiments, the temperature of the decorated glass structure is maintained below 500° C., 400° C., 300° C., 200° C., or 100° C. In a particular embodiment, the decorated glass is maintained at or below room temperature during bending. In a particular embodiment, the decorated glass is not actively heated via a heating element, furnace, oven, etc. during bending, as is the case when hot-forming glass to a curved shape.

(52) As noted above, in addition to providing processing advantages such as eliminating expensive and/or slow heating steps, the cold-forming processes discussed herein are believed to generate curved decorated glasses with a variety of properties that are believed to be superior to those achievable via hot-forming processes. For example, Applicant believes that, for at least some glass materials, heating during hot-forming processes decreases optical properties of curved glass sheets, and thus, the curved decorated glasses formed utilizing the cold-bending processes/systems discussed herein provide for both curved glass shape along with improved optical qualities not believed achievable with hot-bending processes.

(53) Further, many glass coating materials (e.g., anti-glare coatings, anti-reflective coatings, etc.) are applied via deposition processes, such as sputtering processes, that are typically ill-suited for coating on to a curved surface. In addition, many coating materials, such as the polymer layer, also are not able to survive the high temperatures associated with hot-bending processes. Thus, in particular embodiments discussed herein, layer **2020** is applied to glass layer **2010** prior to cold-bending. Thus, Applicant believes that the processes and systems discussed herein allow for bending of glass after one or more coating material has been applied to the glass, in contrast to typical hot-forming processes.

(54) At step **2240**, the curved decorated glass is attached or affixed to the curved support. In various embodiments, the attachment between the curved decorated glass structure and the curved support may be accomplished via an adhesive material. Such adhesives may include any suitable optically clear adhesive for bonding the decorated glass structure in place relative to the display assembly (e.g., to the frame of the display). In one example, the adhesive may include an optically clear adhesive available from 3M Corporation under the trade name 8215. The thickness of the adhesive may be in a range from about 200 μm to about 500 μm.

(55) The adhesive material may be applied in a variety ways. In one embodiment, the adhesive is applied using an applicator gun and made uniform using a roller or a draw down die. In various embodiments, the adhesives discussed herein are structural adhesives. In particular embodiments, the structural adhesives may include an adhesive selected from one or more of the categories: (a) Toughened Epoxy (Masterbond EP21TDCHT-LO, 3M Scotch Weld Epoxy DP460 Off-white); (b) Flexible Epoxy (Masterbond EP21TDC-2LO, 3M Scotch Weld Epoxy 2216 B/A Gray); (c) Acrylic (LORD Adhesive 410/Accelerator 19 w/LORD AP 134 primer, LORD Adhesive 852/LORD Accelerator 25GB, Loctite HF8000, Loctite AA4800); (d) Urethanes (3M Scotch Weld Urethane DP640 Brown); and (e) Silicones (Dow Corning 995). In some cases, structural glues available in sheet format (such as B-staged epoxy adhesives) may be utilized. Furthermore, pressure sensitive structural adhesives such as 3M VHB tapes may be utilized. In such embodiments, utilizing a pressure sensitive adhesive allows for the curved decorated glass to be bonded to the frame without the need for a curing step.

(56) Referring to FIG. **9**, another method **2300** for forming a display utilizing a curved decorated glass structure is shown and described. In some embodiments, the glass layer (e.g., glass layer **2010**) of a decorated glass is formed to curved shape at step **2310**. Shaping at step **2310** may be either cold-forming or hot-forming. At step **2320**, the decorated glass ink/pigment layers and any of the other optional layers are applied to the glass layer following shaping. Next at step **2330**, the curved decorated glass is attached to a frame, such as frame **2110** of display assembly **2100**, or other frame that may be associated with a vehicle interior system.

Glass Materials

(57) The various glass layer(s) of the decorated glass discussed herein, such as glass layer **2010**, may be formed from any suitable glass composition including soda lime glass, aluminosilicate glass, borosilicate glass, boroaluminosilicate glass, alkali-containing aluminosilicate glass, alkali-containing borosilicate glass, and alkali-containing boroaluminosilicate glass.

(58) Unless otherwise specified, the glass compositions disclosed herein are described in mole percent (mol %) as analyzed on an oxide basis.

(59) In one or more embodiments, the glass composition may include SiO_2 in an amount in a range from about 66 mol % to about 80 mol %, from about 67 mol % to about 80 mol %, from about 68 mol % to about 80 mol %, from about 69 mol % to about 80 mol %, from about 70 mol % to about 80 mol %, from about 72 mol % to about 80 mol %, from about 65 mol % to about 78 mol %, from about 65 mol % to about 76 mol %, from about 65 mol % to about 75 mol %, from about 65 mol % to about 74 mol %, from about 65 mol % to about 72 mol %, or from about 65 mol % to about 70 mol %, and all ranges and sub-ranges therebetween.

(60) In one or more embodiments, the glass composition includes Al_2O_3 in an amount greater than about 4 mol %, or greater than about 5 mol %. In one or more embodiments, the glass composition includes Al_2O_3 in a range from greater than about 7 mol % to about 15 mol %, from greater than about 7 mol % to about 14 mol %, from about 7 mol % to about 13 mol %, from about 4 mol % to about 12 mol %, from about 7 mol % to about 11 mol %, from about 8 mol % to about 15 mol %, from 9 mol % to about 15 mol %, from about 9 mol % to about 15 mol %, from about 10 mol % to about 15 mol %, from about 11 mol % to about 15 mol %, or from about 12 mol % to about 15 mol %, and all ranges and sub-ranges therebetween. In one or more embodiments, the upper limit of Al_2O_3 may be about 14 mol %, 14.2 mol %, 14.4 mol %, 14.6 mol %, or 14.8 mol %.

(61) In one or more embodiments, glass layer(s) herein are described as an aluminosilicate glass article or including an aluminosilicate glass composition. In such embodiments, the glass composition or article formed therefrom includes SiO_2 and Al_2O_3 and is not a soda lime silicate glass. In this regard, the glass composition or article formed therefrom includes Al_2O_3 in an amount of about 2 mol % or greater, 2.25 mol % or greater, 2.5 mol % or greater, about 2.75 mol % or greater, about 3 mol % or greater.

(62) In one or more embodiments, the glass composition comprises B_2O_3 (e.g., about 0.01 mol % or greater). In one or more embodiments, the glass composition comprises B_2O_3 in an amount in a range from about 0 mol % to about 5 mol %, from about 0 mol % to about 4 mol %, from about 0 mol % to about 3 mol %, from about 0 mol % to about 2 mol %, from about 0 mol % to about 1 mol %, from about 0 mol % to about 0.5 mol %, from about 0.1 mol % to about 5 mol %, from about 0.1 mol % to about 4 mol %, from about 0.1 mol % to about 3 mol %, from about 0.1 mol % to about 2 mol %, from about 0.1 mol % to about 1 mol %, from about 0.1 mol % to about 0.5 mol %, and all ranges and sub-ranges therebetween. In one or more embodiments, the glass composition is substantially free of B_2O_3.

(63) As used herein, the phrase “substantially free” with respect to the components of the composition means that the component is not actively or intentionally added to the composition during initial batching, but may be present as an impurity in an amount less than about 0.001 mol %.

(64) In one or more embodiments, the glass composition optionally comprises P.sub.2O.sub.5 (e.g., about 0.01 mol % or greater). In one or more embodiments, the glass composition comprises a non-zero amount of P.sub.2O.sub.5 up to and including 2 mol %, 1.5 mol %, 1 mol %, or 0.5 mol %. In one or more embodiments, the glass composition is substantially free of P.sub.2O.sub.5.
(65) In one or more embodiments, the glass composition may include a total amount of R.sub.2O (which is the total amount of alkali metal oxide such as Li.sub.2O, Na.sub.2O, K.sub.2O, Rb.sub.2O, and Cs.sub.2O) that is greater than or equal to about 8 mol %, greater than or equal to about 10 mol %, or greater than or equal to about 12 mol %. In some embodiments, the glass composition includes a total amount of R.sub.2O in a range from about 8 mol % to about 20 mol %, from about 8 mol % to about 18 mol %, from about 8 mol % to about 16 mol %, from about 8 mol % to about 14 mol %, from about 8 mol % to about 12 mol %, from about 9 mol % to about 20 mol %, from about 10 mol % to about 20 mol %, from about 11 mol % to about 20 mol %, from about 12 mol % to about 20 mol %, from about 13 mol % to about 20 mol %, from about 10 mol % to about 14 mol %, or from 11 mol % to about 13 mol %, and all ranges and sub-ranges therebetween. In one or more embodiments, the glass composition may be substantially free of Rb.sub.2O, Cs.sub.2O or both Rb.sub.2O and Cs.sub.2O. In one or more embodiments, the R.sub.2O may include the total amount of Li.sub.2O, Na.sub.2O and K.sub.2O only. In one or more embodiments, the glass composition may comprise at least one alkali metal oxide selected from Li.sub.2O, Na.sub.2O and K.sub.2O, wherein the alkali metal oxide is present in an amount greater than about 8 mol % or greater.
(66) In one or more embodiments, the glass composition comprises Na.sub.2O in an amount greater than or equal to about 8 mol %, greater than or equal to about 10 mol %, or greater than or equal to about 12 mol %. In one or more embodiments, the composition includes Na.sub.2O in a range from about from about 8 mol % to about 20 mol %, from about 8 mol % to about 18 mol %, from about 8 mol % to about 16 mol %, from about 8 mol % to about 14 mol %, from about 8 mol % to about 12 mol %, from about 9 mol % to about 20 mol %, from about 10 mol % to about 20 mol %, from about 11 mol % to about 20 mol %, from about 12 mol % to about 20 mol %, from about 13 mol % to about 20 mol %, from about 10 mol % to about 14 mol %, or from 11 mol % to about 16 mol %, and all ranges and sub-ranges therebetween.
(67) In one or more embodiments, the glass composition includes less than about 4 mol % K.sub.2O, less than about 3 mol % K.sub.2O, or less than about 1 mol % K.sub.2O. In some instances, the glass composition may include K.sub.2O in an amount in a range from about 0 mol % to about 4 mol %, from about 0 mol % to about 3.5 mol %, from about 0 mol % to about 3 mol %, from about 0 mol % to about 2.5 mol %, from about 0 mol % to about 2 mol %, from about 0 mol % to about 1.5 mol %, from about 0 mol % to about 1 mol %, from about 0 mol % to about 0.5 mol %, from about 0 mol % to about 0.2 mol %, from about 0 mol % to about 0.1 mol %, from about 0.5 mol % to about 4 mol %, from about 0.5 mol % to about 3.5 mol %, from about 0.5 mol % to about 3 mol %, from about 0.5 mol % to about 2.5 mol %, from about 0.5 mol % to about 2 mol %, from about 0.5 mol % to about 1.5 mol %, or from about 0.5 mol % to about 1 mol %, and all ranges and sub-ranges therebetween. In one or more embodiments, the glass composition may be substantially free of K.sub.2O.
(68) In one or more embodiments, the glass composition is substantially free of Li.sub.2O.
(69) In one or more embodiments, the amount of Na.sub.2O in the composition may be greater than the amount of Li.sub.2O. In some instances, the amount of Na.sub.2O may be greater than the combined amount of Li.sub.2O and K.sub.2O. In one or more alternative embodiments, the amount of Li.sub.2O in the composition may be greater than the amount of Na.sub.2O or the combined amount of Na.sub.2O and K.sub.2O.
(70) In one or more embodiments, the glass composition may include a total amount of RO (which is the total amount of alkaline earth metal oxide such as CaO, MgO, BaO, ZnO and SrO) in a range from about 0 mol % to about 2 mol %. In some embodiments, the glass composition includes a

non-zero amount of RO up to about 2 mol %. In one or more embodiments, the glass composition comprises RO in an amount from about 0 mol % to about 1.8 mol %, from about 0 mol % to about 1.6 mol %, from about 0 mol % to about 1.5 mol %, from about 0 mol % to about 1.4 mol %, from about 0 mol % to about 1.2 mol %, from about 0 mol % to about 1 mol %, from about 0 mol % to about 0.8 mol %, from about 0 mol % to about 0.5 mol %, and all ranges and sub-ranges therebetween.

(71) In one or more embodiments, the glass composition includes CaO in an amount less than about 1 mol %, less than about 0.8 mol %, or less than about 0.5 mol %. In one or more embodiments, the glass composition is substantially free of CaO. In some embodiments, the glass composition comprises MgO in an amount from about 0 mol % to about 7 mol %, from about 0 mol % to about 6 mol %, from about 0 mol % to about 5 mol %, from about 0 mol % to about 4 mol %, from about 0.1 mol % to about 7 mol %, from about 0.1 mol % to about 6 mol %, from about 0.1 mol % to about 5 mol %, from about 0.1 mol % to about 4 mol %, from about 1 mol % to about 7 mol %, from about 2 mol % to about 6 mol %, or from about 3 mol % to about 6 mol %, and all ranges and sub-ranges therebetween.

(72) In one or more embodiments, the glass composition comprises ZrO_2 in an amount equal to or less than about 0.2 mol %, less than about 0.18 mol %, less than about 0.16 mol %, less than about 0.15 mol %, less than about 0.14 mol %, less than about 0.12 mol %. In one or more embodiments, the glass composition comprises ZrO_2 in a range from about 0.01 mol % to about 0.2 mol %, from about 0.01 mol % to about 0.18 mol %, from about 0.01 mol % to about 0.16 mol %, from about 0.01 mol % to about 0.15 mol %, from about 0.01 mol % to about 0.14 mol %, from about 0.01 mol % to about 0.12 mol %, or from about 0.01 mol % to about 0.10 mol %, and all ranges and sub-ranges therebetween.

(73) In one or more embodiments, the glass composition comprises SnO_2 in an amount equal to or less than about 0.2 mol %, less than about 0.18 mol %, less than about 0.16 mol %, less than about 0.15 mol %, less than about 0.14 mol %, less than about 0.12 mol %. In one or more embodiments, the glass composition comprises SnO_2 in a range from about 0.01 mol % to about 0.2 mol %, from about 0.01 mol % to about 0.18 mol %, from about 0.01 mol % to about 0.16 mol %, from about 0.01 mol % to about 0.15 mol %, from about 0.01 mol % to about 0.14 mol %, from about 0.01 mol % to about 0.12 mol %, or from about 0.01 mol % to about 0.10 mol %, and all ranges and sub-ranges therebetween.

(74) In one or more embodiments, the glass composition may include an oxide that imparts a color or tint to the glass articles. In some embodiments, the glass composition includes an oxide that prevents discoloration of the glass article when the glass article is exposed to ultraviolet radiation. Examples of such oxides include, without limitation oxides of: Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Ce, W, and Mo.

(75) In one or more embodiments, the glass composition includes Fe expressed as Fe_2O_3, wherein Fe is present in an amount up to (and including) about 1 mol %. In some embodiments, the glass composition is substantially free of Fe. In one or more embodiments, the glass composition comprises Fe_2O_3 in an amount equal to or less than about 0.2 mol %, less than about 0.18 mol %, less than about 0.16 mol %, less than about 0.15 mol %, less than about 0.14 mol %, less than about 0.12 mol %. In one or more embodiments, the glass composition comprises Fe_2O_3 in a range from about 0.01 mol % to about 0.2 mol %, from about 0.01 mol % to about 0.18 mol %, from about 0.01 mol % to about 0.16 mol %, from about 0.01 mol % to about 0.15 mol %, from about 0.01 mol % to about 0.14 mol %, from about 0.01 mol % to about 0.12 mol %, or from about 0.01 mol % to about 0.10 mol %, and all ranges and sub-ranges therebetween.

(76) Where the glass composition includes TiO_2, TiO_2 may be present in an amount of about 5 mol % or less, about 2.5 mol % or less, about 2 mol % or less or about 1 mol % or less. In one or more embodiments, the glass composition may be substantially free of TiO_2.

(77) An exemplary glass composition includes SiO_2 in an amount in a range from about 65 mol % to about 75 mol %, Al_2O_3 in an amount in a range from about 8 mol % to about 14 mol %, Na_2O in an amount in a range from about 12 mol % to about 17 mol %, K_2O in an amount in a range of about 0 mol % to about 0.2 mol %, and MgO in an amount in a range from about 1.5 mol % to about 6 mol %. Optionally, SnO_2 may be included in the amounts otherwise disclosed herein.

Strengthened Glass Properties

(78) In one or more embodiments, glass layer **2010** or other glass layer of any of the decorated glass embodiments discussed herein may be formed from a strengthened glass sheet or article. In one or more embodiments, the glass articles used to form the layer(s) of the decorated glass structures discussed herein may be strengthened to include compressive stress that extends from a surface to a depth of compression (DOC). The compressive stress regions are balanced by a central portion exhibiting a tensile stress. At the DOC, the stress crosses from a positive (compressive) stress to a negative (tensile) stress.

(79) In one or more embodiments, the glass articles used to form the layer(s) of the decorated glass structures discussed herein may be strengthened mechanically by utilizing a mismatch of the coefficient of thermal expansion between portions of the glass to create a compressive stress region and a central region exhibiting a tensile stress. In some embodiments, the glass article may be strengthened thermally by heating the glass to a temperature above the glass transition point and then rapidly quenching.

(80) In one or more embodiments, the glass articles used to form the layer(s) of the decorated glass structures discussed herein may be chemically strengthening by ion exchange. In the ion exchange process, ions at or near the surface of the glass article are replaced by—or exchanged with—larger ions having the same valence or oxidation state. In those embodiments in which the glass article comprises an alkali aluminosilicate glass, ions in the surface layer of the article and the larger ions are monovalent alkali metal cations, such as Li^+, Na^+, K^+, Rb^+, and Cs^+. Alternatively, monovalent cations in the surface layer may be replaced with monovalent cations other than alkali metal cations, such as Ag^+ or the like. In such embodiments, the monovalent ions (or cations) exchanged into the glass article generate a stress.

(81) Ion exchange processes are typically carried out by immersing a glass article in a molten salt bath (or two or more molten salt baths) containing the larger ions to be exchanged with the smaller ions in the glass article. It should be noted that aqueous salt baths may also be utilized. In addition, the composition of the bath(s) may include more than one type of larger ion (e.g., Na^+ and K^+) or a single larger ion. It will be appreciated by those skilled in the art that parameters for the ion exchange process, including, but not limited to, bath composition and temperature, immersion time, the number of immersions of the glass article in a salt bath (or baths), use of multiple salt baths, additional steps such as annealing, washing, and the like, are generally determined by the composition of the glass layer(s) of a decorated glass structure (including the structure of the article and any crystalline phases present) and the desired DOC and CS of the glass layer(s) of a decorated glass structure that results from strengthening.

(82) Exemplary molten bath composition may include nitrates, sulfates, and chlorides of the larger alkali metal ion. Typical nitrates include KNO_3, $NaNO_3$, $LiNO_3$, $NaSO_4$ and combinations thereof. The temperature of the molten salt bath typically is in a range from about 380° C. up to about 450° C., while immersion times range from about 15 minutes up to about 100 hours depending on the glass thickness, bath temperature and glass (or monovalent ion) diffusivity. However, temperatures and immersion times different from those described above may also be used.

(83) In one or more embodiments, the glass articles used to form the layer(s) of the decorated glass may be immersed in a molten salt bath of 100% $NaNO_3$, 100% KNO_3, or a combination of $NaNO_3$ and KNO_3 having a temperature from about 370° C. to about 480° C. In some

embodiments, the glass layer(s) of a decorated glass may be immersed in a molten mixed salt bath including from about 5% to about 90% KNO.sub.3 and from about 10% to about 95% NaNO.sub.3. In one or more embodiments, the glass article may be immersed in a second bath, after immersion in a first bath. The first and second baths may have different compositions and/or temperatures from one another. The immersion times in the first and second baths may vary. For example, immersion in the first bath may be longer than the immersion in the second bath.
(84) In one or more embodiments, the glass articles used to form the layer(s) of the decorated glass structures may be immersed in a molten, mixed salt bath including NaNO.sub.3 and KNO.sub.3 (e.g., 49%/51%, 50%/50%, 51%/49%) having a temperature less than about 420° C. (e.g., about 400° C. or about 380° C.). for less than about 5 hours, or even about 4 hours or less.
(85) Ion exchange conditions can be tailored to provide a “spike” or to increase the slope of the stress profile at or near the surface of the resulting glass layer(s) of a decorated glass structure. The spike may result in a greater surface CS value. This spike can be achieved by single bath or multiple baths, with the bath(s) having a single composition or mixed composition, due to the unique properties of the glass compositions used in the glass layer(s) of a decorated glass structure described herein.
(86) In one or more embodiments, where more than one monovalent ion is exchanged into the glass articles used to form the layer(s) of the decorated glass structures, the different monovalent ions may exchange to different depths within the glass layer (and generate different magnitudes stresses within the glass article at different depths). The resulting relative depths of the stress-generating ions can be determined and cause different characteristics of the stress profile.
(87) CS is measured using those means known in the art, such as by surface stress meter (FSM) using commercially available instruments such as the FSM-6000, manufactured by Orihara Industrial Co., Ltd. (Japan). Surface stress measurements rely upon the accurate measurement of the stress optical coefficient (SOC), which is related to the birefringence of the glass. SOC in turn is measured by those methods that are known in the art, such as fiber and four point bend methods, both of which are described in ASTM standard C770-98 (2013), entitled “Standard Test Method for Measurement of Glass Stress-Optical Coefficient,” the contents of which are incorporated herein by reference in their entirety, and a bulk cylinder method. As used herein CS may be the “maximum compressive stress” which is the highest compressive stress value measured within the compressive stress layer. In some embodiments, the maximum compressive stress is located at the surface of the glass article. In other embodiments, the maximum compressive stress may occur at a depth below the surface, giving the compressive profile the appearance of a “buried peak.”
(88) DOC may be measured by FSM or by a scattered light polariscope (SCALP) (such as the SCALP-04 scattered light polariscope available from Glasstress Ltd., located in Tallinn Estonia), depending on the strengthening method and conditions. When the glass article is chemically strengthened by an ion exchange treatment, FSM or SCALP may be used depending on which ion is exchanged into the glass article. Where the stress in the glass article is generated by exchanging potassium ions into the glass article, FSM is used to measure DOC. Where the stress is generated by exchanging sodium ions into the glass article, SCALP is used to measure DOC. Where the stress in the glass article is generated by exchanging both potassium and sodium ions into the glass, the DOC is measured by SCALP, since it is believed the exchange depth of sodium indicates the DOC and the exchange depth of potassium ions indicates a change in the magnitude of the compressive stress (but not the change in stress from compressive to tensile); the exchange depth of potassium ions in such glass articles is measured by FSM. Central tension or CT is the maximum tensile stress and is measured by SCALP.
(89) In one or more embodiments, the glass articles used to form the layer(s) of the decorated glass structures may be strengthened to exhibit a DOC that is described a fraction of the thickness t of the glass article (as described herein). For example, in one or more embodiments, the DOC may be equal to or greater than about 0.05 t, equal to or greater than about 0.1 t, equal to or greater than

about 0.11 t, equal to or greater than about 0.12 t, equal to or greater than about 0.13 t, equal to or greater than about 0.14 t, equal to or greater than about 0.15 t, equal to or greater than about 0.16 t, equal to or greater than about 0.17 t, equal to or greater than about 0.18 t, equal to or greater than about 0.19 t, equal to or greater than about 0.2 t, equal to or greater than about 0.21 t. In some embodiments, The DOC may be in a range from about 0.08 t to about 0.25 t, from about 0.09 t to about 0.25 t, from about 0.18 t to about 0.25 t, from about 0.11 t to about 0.25 t, from about 0.12 t to about 0.25 t, from about 0.13 t to about 0.25 t, from about 0.14 t to about 0.25 t, from about 0.15 t to about 0.25 t, from about 0.08 t to about 0.24 t, from about 0.08 t to about 0.23 t, from about 0.08 t to about 0.22 t, from about 0.08 t to about 0.21 t, from about 0.08 t to about 0.2 t, from about 0.08 t to about 0.19 t, from about 0.08 t to about 0.18 t, from about 0.08 t to about 0.17 t, from about 0.08 t to about 0.16 t, or from about 0.08 t to about 0.15 t. In some instances, the DOC may be about 20 μm or less. In one or more embodiments, the DOC may be about 40 μm or greater (e.g., from about 40 μm to about 300 μm, from about 50 μm to about 300 μm, from about 60 μm to about 300 μm, from about 70 μm to about 300 μm, from about 80 μm to about 300 μm, from about 90 μm to about 300 μm, from about 100 μm to about 300 μm, from about 110 μm to about 300 μm, from about 120 μm to about 300 μm, from about 140 μm to about 300 μm, from about 150 μm to about 300 μm, from about 40 μm to about 290 μm, from about 40 μm to about 280 μm, from about 40 μm to about 260 μm, from about 40 μm to about 250 μm, from about 40 μm to about 240 μm, from about 40 μm to about 230 μm, from about 40 μm to about 220 μm, from about 40 μm to about 210 μm, from about 40 μm to about 200 μm, from about 40 μm to about 180 μm, from about 40 μm to about 160 μm, from about 40 μm to about 150 μm, from about 40 μm to about 140 μm, from about 40 μm to about 130 μm, from about 40 μm to about 120 μm, from about 40 μm to about 110 μm, or from about 40 μm to about 100 μm.

(90) In one or more embodiments, the glass articles used to form the layer(s) of the decorated glass structures may have a CS (which may be found at the surface or a depth within the glass article) of about 200 MPa or greater, 300 MPa or greater, 400 MPa or greater, about 500 MPa or greater, about 600 MPa or greater, about 700 MPa or greater, about 800 MPa or greater, about 900 MPa or greater, about 930 MPa or greater, about 1000 MPa or greater, or about 1050 MPa or greater.

(91) In one or more embodiments, the glass articles used to form the layer(s) of the decorated glass structures may have a maximum tensile stress or central tension (CT) of about 20 MPa or greater, about 30 MPa or greater, about 40 MPa or greater, about 45 MPa or greater, about 50 MPa or greater, about 60 MPa or greater, about 70 MPa or greater, about 75 MPa or greater, about 80 MPa or greater, or about 85 MPa or greater. In some embodiments, the maximum tensile stress or central tension (CT) may be in a range from about 40 MPa to about 100 MPa.

(92) Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order. Accordingly, where a method claim does not actually recite an order to be followed by its steps or it is not otherwise specifically stated in the claims or descriptions that the steps are to be limited to a specific order, it is in no way intended that any particular order be inferred. In addition, as used herein, the article “a” is intended to include one or more than one component or element, and is not intended to be construed as meaning only one.

(93) It will be apparent to those skilled in the art that various modifications and variations can be made without departing from the spirit or scope of the disclosed embodiments. Since modifications, combinations, sub-combinations and variations of the disclosed embodiments incorporating the spirit and substance of the embodiments may occur to persons skilled in the art, the disclosed embodiments should be construed to include everything within the scope of the appended claims and their equivalents.

Claims

1. A decorated glass, comprising: a transparent substrate having a first major surface and a second major surface, the second major surface being opposite the first major surface; and a black ink layer disposed on the second major surface in a display region, the black ink layer having a transmission coefficient of between 0.2 and 0.85 with respect to incident light having a wavelength of 400 nm to 700 nm; wherein the decorated glass has 2% or less of sparkle when measured from the first major surface via pixel power deviation reference (PPDr).
2. The decorated glass of claim 1, wherein the transparent substrate comprises at least one of soda lime glass, aluminosilicate glass, borosilicate glass, boroaluminosilicate glass, alkali-containing aluminosilicate glass, alkali-containing borosilicate glass, or alkali-containing boroaluminosilicate glass.
3. The decorated glass of claim 1, wherein the black ink layer has a thickness of at least 0.05 μm and no more than 50 μm.
4. The decorated glass of claim 1, wherein the black ink layer comprises K ink or LK ink from the CMYK color model.
5. The decorated glass of claim 1, wherein the black ink layer comprises CMYK composite black ink according to the CMYK color model.
6. The decorated glass of claim 1, wherein the black ink layer comprises CMYK rich black.
7. The decorated glass of claim 1, further comprising an opaque mask layer, the opaque mask layer defining at least one non-display region in which the decorated glass transmits at most 5% of incident light.
8. The decorated glass of claim 7, wherein the opaque mask layer is disposed between the second major surface and the black ink layer.
9. The decorated glass of claim 7, wherein the black ink layer is disposed between the second major surface and the opaque mask layer.
10. The decorated glass of claim 1, wherein the ink layer has a transmission coefficient of 0.5 to 0.85 with respect to the incident light having a wavelength of 400 nm to 700 nm and wherein the black ink layer provides a color matching effect when the decorated glass is placed over a light source.
11. The decorated glass of claim 1, wherein the ink layer has a transmission coefficient of 0.2 to 0.7 with respect to the incident light having a wavelength of 400 nm to 700 nm and wherein the black ink layer provides a deadfronting effect when the decorated glass is placed over a light source.
12. A device, comprising: a decorated glass having a first side and a second side, the second side being opposite the first side, the decorated glass comprising: a transparent substrate having a first major surface and a second major surface, the second major surface being opposite the first major surface; and a black ink layer disposed on the second major surface in at least one display region, wherein the ink layer has a transmission coefficient of between 0.2 and 0.85 with respect to incident light having a wavelength of 400 nm to 700 nm; and a light source disposed on the second side of the decorated glass; wherein 2% or less of sparkle is measured from the first side of the decorated glass via pixel power deviation reference (PPDr).
13. The device according to claim 12, wherein the light source is at least one of a light emitting diode (LED) display, an organic LED (OLED) display, a liquid crystal display (LCD), or a plasma display.
14. The device according to claim 12, wherein the transparent substrate comprises at least one of soda lime glass, aluminosilicate glass, borosilicate glass, boroaluminosilicate glass, alkali-containing aluminosilicate glass, alkali-containing borosilicate glass, or alkali-containing boroaluminosilicate glass.
15. The device according to claim 12, wherein the black ink layer has a thickness of at least 0.05 μm and no more than 50 μm.
16. The device according to claim 12, wherein the black ink layer comprises K ink from the

CMYK color model.

17. The device according to claim 12, wherein the black ink layer comprises CMYK composite black ink according to the CMYK color model.

18. The device according to claim 12, wherein the black ink layer comprised CMYK rich black according to the CMYK color model.

19. The device according to claim 12, further comprising an opaque mask layer, the opaque mask layer defining at least one non-display region in which the decorated glass transmits at most 5% of incident light.